Chapter 2: Unit Conversions

Math is a Requirement

- *All science requires mathematics. The knowledge of mathematical things is almost innate in us. . . [Mathematics] is the easiest of sciences, a fact which is obvious in that no one's brain rejects it…*
  - Roger Bacon (c. 1214-c. 1294)

- *Stand firm in your refusal to remain conscious during algebra. In real life, I assure you, there is no such thing as algebra.*
  - Fran Lebowitz (b. 1951)

Significant Figures

- Meaningful Digits in reported values that express the precision of the value.
- **Nonzero integers** - Always count as significant figures
  - 3456 has four significant figures

Rules for Counting Significant Figures

- Zeros - There are three classes of zeros
  - **Leading zeros** - Zeros that precede all of the nonzero digits and are never counted as significant figures
    - 0.048 has two significant figures
  - **Captive zeros** - Zeros that fall between nonzero digits and always count as significant figures
    - 16.07 has four significant figures

Rules for Counting Significant Figures

- **Trailing zeros** - Zeros at the right end of the number
  - Significant only if the number contains a decimal point
    - 9.300 has four significant figures
    - 150 has two significant figures
- **Exact numbers** - Possess an unlimited number of significant figures
  - 1 inch = 2.54 cm
  - 9 pencils
    - Obtained by counting

Exponential Notation or Scientific Notation

- Example
  - 300 written as 3.00×10^2
  - Contains three significant figures
- Advantages
  - Number of significant figures can be easily indicated
  - Fewer zeros are needed to write a very large or very small number

Rules for Rounding

- Rule 1 - If the digit to be removed is less than 5, the preceding digit stays the same
  - 5.64 rounds to 5.6
    - If the final result has two significant figures
- If the digit to be removed is equal to or greater than 5, the preceding digit is increased by 1
  - 5.68 rounds to 5.7
    - If the final result has two significant figures
  - 3.861 rounds to 3.9
    - If the final result has two significant figures

Rules for Rounding

- Rule 2 - In a series of calculations, carry the extra digits through to the final result and then round off
  - This means that you should carry all of the digits that show on your calculator until you arrive at the final number
  - Then, round off using the procedures in Rule 1

Significant Figures in Mathematical Operations

- For **multiplication or division**, the number of significant figures in the result is the same as that in the measurement with the smallest number of significant figures

$$1.342 \times \underline{5.5} = 7.381 \rightarrow \underline{7.4}$$

Significant Figures in Mathematical Operations

- For **addition or subtraction**, the limiting term is the one with the smallest number of decimal places

$$
\begin{array}{rr}
 & 23.445 \\
+ & 7.83 \\
\hline
 & 31.2\underline{7}5
\end{array}
\xrightarrow{\text{Corrected}} 31.2\underline{8}
$$

Dimensional Analysis

- Mathematical Process that allows for the ability to convert a value containing one type of unit to a value having a different type of unit.
  - Ex.
    - 3.5 tsp = ? mL
- **Equivalency Statement** – provides a relationship between two different units.
- **Conversion Factor** – A ratio formed from an equivalency statement to describe the relationship between two units as a fraction.
  - 1tsp = 5 mL

Metric – Metric Conversions

- The relationship between metric units can be derived from the metric prefixes.

Prefixes for units larger than the base unit			Prefixes for units smaller than the base unit		
Prefix	Abbreviation	Definition	Prefix	Abbreviation	Definition
giga	G	1,000,000,000 or 10^9	centi	c	0.01 or 10^{-2}
mega	M	1,000,000 or 10^6	milli	m	0.001 or 10^{-3}
kilo	k	1000 or 10^3	micro	μ	0.000001 or 10^{-6}
			nano	n	0.000000001 or 10^{-9}
			pico	p	0.000000000001 or 10^{-12}

Metric – Metric Conversions

- **One-step Conversions**
  - 453.94 L = ? GL
- **Two-step Conversions**
  - 12.7 nm = ? Km

English – Metric Conversions

Type of measurement	Probably most useful to know	Also useful to know
length	$\frac{2.54\ \text{cm}}{1\ \text{in.}}$ (exact)	$\frac{1.609\ \text{km}}{1\ \text{mi}}$ $\frac{39.37\ \text{in.}}{1\ \text{m}}$ $\frac{1.094\ \text{yd}}{1\ \text{m}}$
mass	$\frac{453.6\ \text{g}}{1\ \text{lb}}$	$\frac{2.205\ \text{lb}}{1\ \text{kg}}$
volume	$\frac{3.785\ \text{L}}{1\ \text{gal}}$	$\frac{1.057\ \text{qt}}{1\ \text{L}}$

English – Metric Conversions

- Conversions with **fractional units**
  - 45.8 mi/hr = ? Km/s
- Conversions with **exponential units**
  - 23.4 in^3 = ? mL

Density

- Defined:
  - $density = \frac{mass}{Volume}$ OR $d = \frac{m}{V}$
    - Density of lead: 11.34 g/cm^3
    - Density of water: 1.00 g/mL

Density

- Densities for liquids and solids generally decrease when temperature is lowered.
  - Many densities are recorded with the temperature at which they are measured
    - Density of Ethanol:
      - 0.806 g/mL at 0°C
      - 0.789 g/mL at 20°C

Density Calculations

- What is the density of sample of platinum that has a mass of 4.30 g and a volume of 2.00 cm^3?

- A sample of copper has a mass of 44.291 g. A graduated cylinder was used to determine the volume of the copper sample. The initial volume of water in the graduated cylinder was 22.3 mL. After the copper sample was placed in the graduated cylinder the volume of water rose to 27.2 mL. What is the density of this copper sample?

Density Calculations

- The density of lead is 11.34 g/cm^3. What is the volume of a sample of lead that has a mass of 18.236 g in kL?

Percentage By Mass

- The number of mass units for the part for each 100 mass units of the whole.

$$X\%\ by\ mass = \frac{X\ (any\ mass\ unit) part}{100\ (\ same\ mass\ unit) whole}$$

$$X\%\ by\ volume = \frac{X\ (any\ volume\ unit) part}{100\ (\ same\ volume\ unit) whole}$$

Percentage By Mass

- The mass of the ocean is about 1.8 × 1021 kg. If the ocean contains 0.014% by mass hydrogen carbonate ions, HCO3-, how many pounds of HCO3- are in the ocean?

- When you are doing heavy work, your muscles get about 75 to 80% by volume of your blood. If your body contains 5.2 liters of blood, how many liters of blood are in your muscles when you are working hard enough to send them 78% by volume of your blood?

Steps to Unit Conversions

- **Step 1: State your question in an expression that sets the unknown unit(s) equal to one or more of the values given.**
  - To the left of the equals sign, show the unit(s) you want in your answer.
  - To the right of the equals sign, start with an expression composed of the given unit(s) that parallels in kind and placement the units you want in your answer.
    - If you want a single unit in your answer, start with a value that has a single unit.
    - If you want a ratio of two units in your answer, start with a value that has a ratio of two units, or start with a ratio of two values, each of which has one unit. Put each type of unit in the position you want it to have in the answer.

Steps to Unit Conversions

- **Step 2: Multiply the expression to the right of the equals sign by conversion factors that cancel unwanted units and generate the desired units.**
  - If you are not certain which conversion factor to use, ask yourself, "What is the fundamental conversion the problem requires, and what conversion factor do I need to make that type of conversion?"

Steps to Unit Conversions

- **Step 3: Do a quick check to be sure you used the correct conversion factors and that your units cancel to yield the desired unit(s).**
- **Step 4: Do the calculation, rounding your answer to the correct number of significant figures and combining it with the correct unit.**